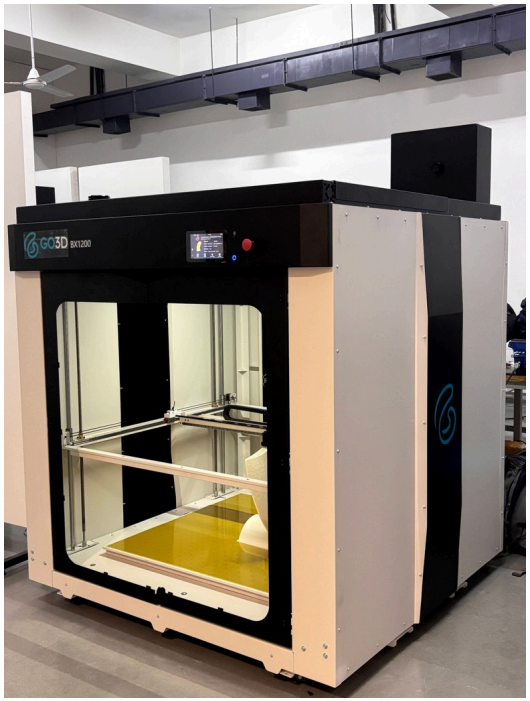
ENGINEERING PORTFOLIO · 2023 — 2026

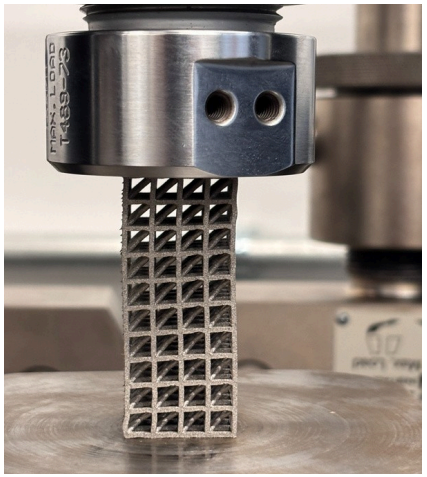
Vedant Nadoda

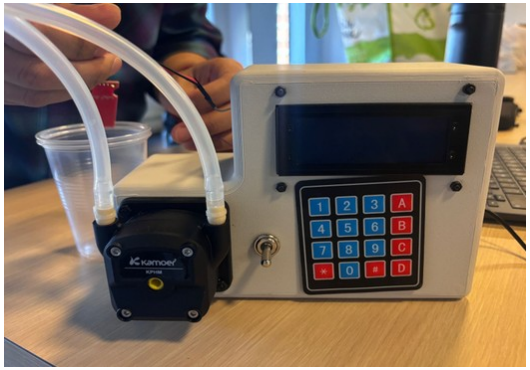
Mechanical Engineer

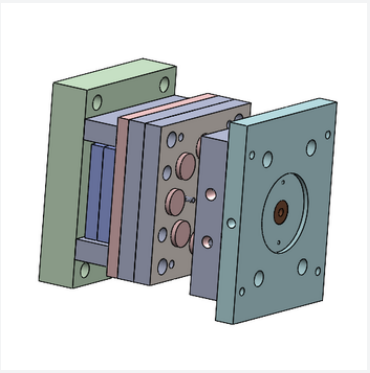
ADDITIVE MANUFACTURING · MECHATRONICS · MANUFACTURING

Hands-on engineer who designs, builds, tests and troubleshoots physical systems — from industrial large-format 3D printers to closed-loop mechatronic prototypes and metal-AM test specimens.









P01 Large-format FDM printer build & commissioning

P02 Closed-loop peristaltic pump, STM32 control

P03 DMLS 316L lattice under UTM compression

P05 Eight-cavity injection mold, SolidWorks

EDUCATION
M.S. Mechanical Engineering
New York University — Dec 2026

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LINKEDIN
linkedin.com/in/vedant-nadoda

LOCATION
New York City Metro Area

WEBSITE
vedant030301.github.io

GITHUB
github.com/Vedant030301

ENGINEERING PROFILE

Design it. Build it. Test it. Find out why it failed — and fix it.

Mechanical engineer with 2+ years of hands-on experience building industrial FDM printers, prototyping mechanisms and testing materials across metal (DMLS/LPBF, WAAM) and polymer (FDM, SLA) platforms. My work spans the full loop: CAD and DFM/DFA up front; assembly, calibration and commissioning on the floor; mechanical testing and root-cause analysis when results don't match the spec.

— AT A GLANCE

67% ↓
Printer assembly lead time
3 days → 1 day · GO3D (P01)

7
Client installations supported
Setup, calibration, acceptance · GO3D (P01)

1200mm
Largest FDM build volume built
Also 500 mm and 300 mm · GO3D (P01)

150+
Students supported in materials labs
SEM, XRD, tensile, hardness · NYU

— EDUCATION & CERTIFICATION

M.S. Mechanical Engineering — New York University — JAN 2025 – DEC 2026

B.Tech Mechanical Engineering — Indus University, India — 2020 – 2023

Advance Certification — Machining Technology & Plastic Engineering — JUN 2024
International Automobile Centre of Excellence (JV of Govt. of Gujarat & Maruti Suzuki) · Graded "Competent"

— WHERE TO SEE IT

01	02	03	04	05	06
Design	**Build**	**Test**	**Analyze**	**Troubleshoot**	**Improve**
P01 P04 P05	P01 P02 P07	P02 P03	P03 P07	P01 P02	P01

SELECTED WORK

P01 **Industrial Large-Format FDM Printers** 03
ADDITIVE MFG · ASSEMBLY · CALIBRATION · INDUSTRY

P02 **Closed-Loop Peristaltic Pump** 05
MECHATRONICS · STM32 · STEPPER CONTROL

P03 **DMLS 316L Stainless-Steel Lattices** 07
METAL AM · MECHANICAL TESTING · FAILURE ANALYSIS

P04 **Hydraulic Lathe Cross-Slide** 09
MECHANISM DESIGN · HYDRAULICS · RETROFIT

P05 **Injection Mold Design** 10
PLASTICS · TOOLING · SOLIDWORKS

P06 **CNC / VMC Machining** 11
G-CODE · TOOLPATHS · METROLOGY

P07 **Rapid Build & Simulation** 12
CNC PLOTTER (1ST PLACE) · COUETTE-FLOW SOLVER

— **Experience & Technical Capabilities** 13
SKILLS MAPPED TO THE PROJECTS THAT USED THEM

TARGET ROLES — FULL-TIME FROM DEC 2026

Mechanical Engineering · Additive Manufacturing · Manufacturing Engineering · Product Development · R&D / Prototyping · Mechatronics · Applications Engineering

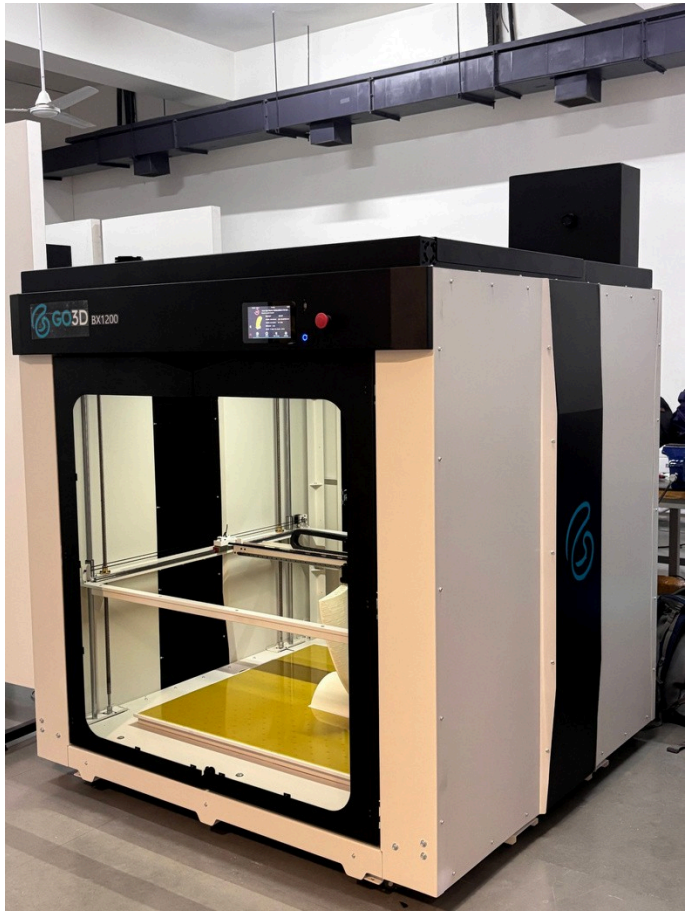


FIG 1.1 Enclosed large-format FDM platform — the class of machine assembled, calibrated and quality-validated before customer handover.

P01 · INDUSTRY · GO3D INNOVATION LLP

Industrial Large-Format FDM Printers — Build, Calibration & Commissioning

Built, calibrated and commissioned custom FDM printers up to 1200 mm build volume, then carried them through customer installation, operator training and support.

ROLE	DURATION	TOOLS
Production Support Engineer Mech. Engineering Intern (prior)	Nov 2023 – Dec 2024 Gandhinagar, India	SolidWorks · Cura · GD&T Lean · Quality inspection

01 PROBLEM / OBJECTIVE

Clients ordered printers to custom specifications at **1200, 500 and 300 mm** build volumes. Each machine had to be assembled, calibrated and validated to spec, then installed and supported on site — and assembly took **3 days per machine**.

02 ENGINEERING APPROACH — PRODUCT LIFECYCLE I SUPPORTED

IN THE FACTORY

01 **Assemble** — mechanical build, modifications

02 **Calibrate** — hardware–software integration

03 **Test & QA** — print-quality validation

AT THE CUSTOMER

04 **Install** — site setup, final acceptance

05 **Train** — operators, SOPs, maintenance

06 **Support** — + custom part production

Design input — SolidWorks
components · fixtures · cable mgmt · mounts

Feedback — root-cause analysis
mechanical & process deviations traced

Process improvement — Lean · 5S · Kaizen · standardized workflows
assembly, inventory, machine organization, BOM accuracy

3 days → 1 day

— BUILD VOLUMES DELIVERED TO CLIENT SPEC

300 mm	500 mm	1200 mm

03 DESIGN & DEVELOPMENT

- **Designed and validated** custom machine components, cable-management systems, fixtures and mounting hardware in SolidWorks, applying DFM/DFA and GD&T.
- **Designed and ran** iterative DOE-style test plans across FDM and resin print parameters to find process limits, quantify print accuracy, and improve calibration, slicing and post-processing (intern phase).

04 BUILD / IMPLEMENTATION

- **Built, assembled and commissioned** printers hands-on — assembly, disassembly, prototype modifications, calibration and testing — to client specifications.
- **Supported on-site installations:** equipment setup, site requirements, calibration, troubleshooting and final acceptance.

05 TESTING & ITERATION

- **Calibrated and validated** print quality; integrated hardware and software.
- **Performed root-cause analysis** on mechanical and process deviations.
- **Wrote SOPs** and trained customers and operators on operation, maintenance, calibration and troubleshooting.

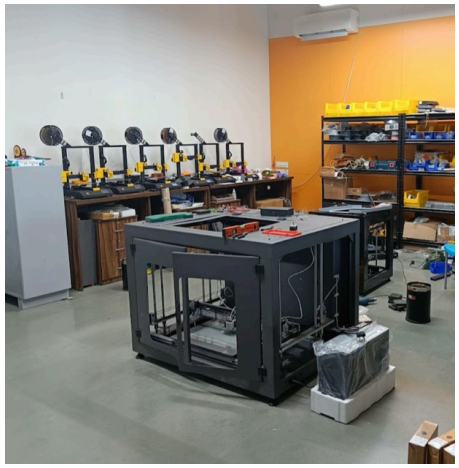


FIG 1.2 Production floor — enclosed large-format frames mid-assembly alongside smaller-format printers.



FIG 1.3 Printed parts with fine relief, thin walls and tall features — geometry that exposes calibration and process errors.

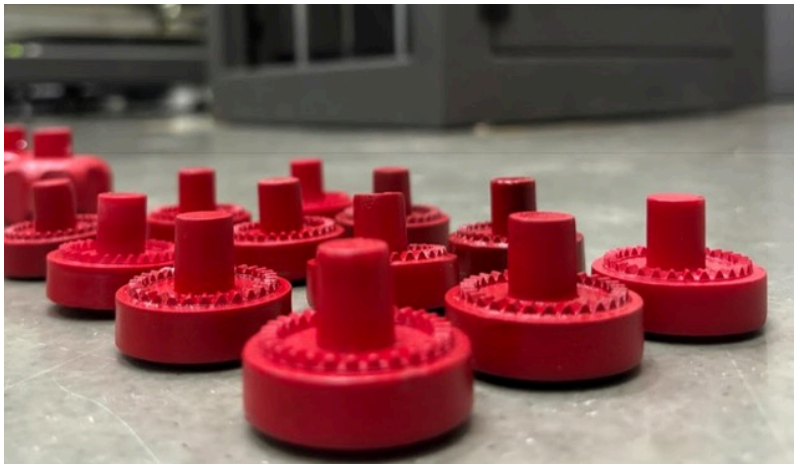


FIG 1.4 Customer-specific components printed in batch — repeatability across a run is the quality target.

06 RESULTS

67% ↓
Assembly lead time, 3 days → 1 day
Standardized, Lean-based workflow (team effort)

98%
Dimensional accuracy
Root-cause & process work contributed

7
Client installations supported
Setup through final acceptance

3
Industry exhibitions
Live demos & client engagement

07 ENGINEERING TAKEAWAY

Production-floor engineering: taking a machine from parts to a calibrated, validated product; tracing the root cause when prints go out of spec; and improving the process that builds the machine — not just the machine itself.

P02 · NEW YORK UNIVERSITY

Closed-Loop Peristaltic Pump with Water-Level Control

Developed an automated peristaltic pumping system for electronically controlled fluid transfer and water-level regulation.

ROLE	DURATION	TOOLS
Control logic, electrical architecture, system integration	Jan – May 2026	STM32 H5 Nucleo · TMC2209 · Stepper · LCD · Keypad · 3D printing · Laser cutting

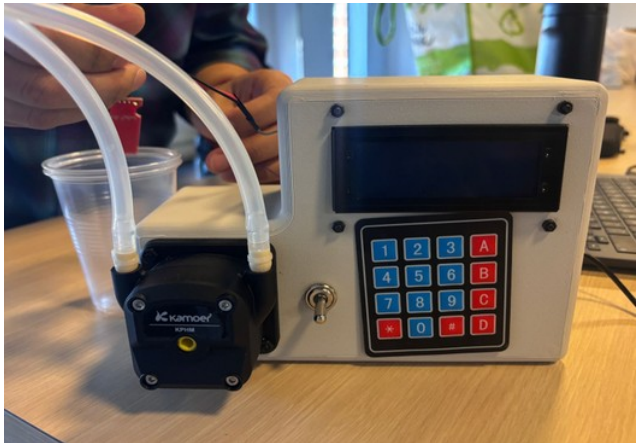
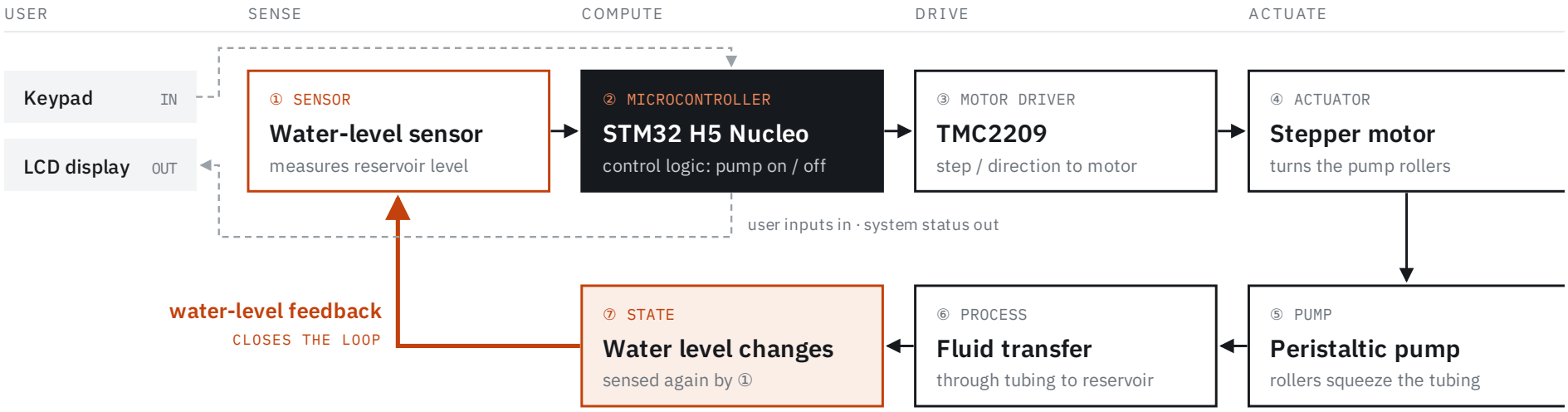


FIG 2.1 Working prototype: stepper-driven peristaltic pump head, enclosure with LCD and 4×4 keypad, emergency switch, and tubing to the reservoir.

01 PROBLEM / OBJECTIVE

Hold a target water level without manual pump operation. The system has to **sense** the level, **decide** when to pump, **drive** a peristaltic pump, and let a user **monitor and set** it — all in one self-contained unit.

02 ENGINEERING APPROACH — CLOSED-LOOP SYSTEM ARCHITECTURE



POWER & SAFETY

24 V DC supply → **Emergency switch** → **DC-DC step-down conversion** → separate voltage levels for motor, display and control

03 DESIGN & DEVELOPMENT

CONTROL

Implemented closed-loop logic on the STM32: water-level feedback decides when the pump actuates.

ACTUATION

Drove a stepper-powered peristaltic pump through a TMC2209 driver. In a peristaltic pump only the tubing touches the fluid.

POWER & SAFETY

Designed the electrical architecture around a 24 V supply with DC-DC conversion for each subsystem's voltage; added an emergency switch.

INTERFACE

Integrated an LCD for system status and a keypad for user input.

05 TESTING & ITERATION

Verified end-to-end operation: level feedback triggered pump actuation automatically, and the LCD/keypad interface worked as intended.

> **LIMITATION FOUND → NEXT ITERATION**
>
> The water-level sensor showed **corrosion and reliability issues** in use. A v2 should prioritize level sensing suited to long-term water exposure — the weakest link in the loop.

04 BUILD / IMPLEMENTATION — WIRING ARCHITECTURE

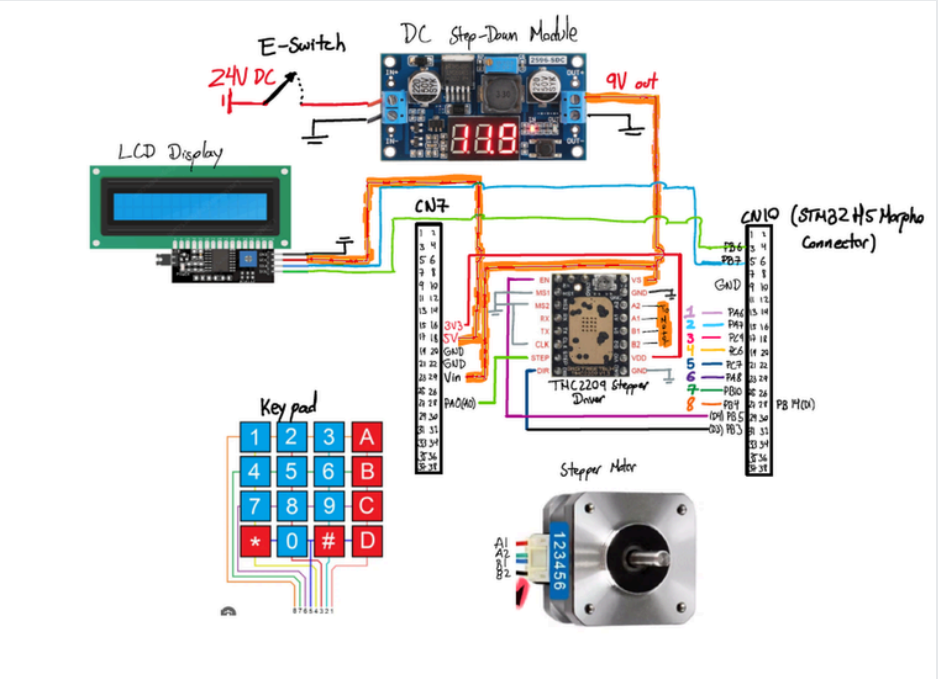


FIG 2.2 Wiring map from supply to actuator: 24 V input, E-switch, DC step-down module, TMC2209 driver, stepper, LCD and 4×4 keypad, pinned to the STM32 H5 Morpho connectors (CN7 / CN10).

- **Integrated** tubing, pump mechanism, level sensor, electronics and fluid reservoirs into a single physical prototype.
- **Fabricated** structural components with 3D printing and laser cutting.

06 RESULTS

Closed-loop regulation

Water level held automatically through pump actuation

Autonomous transfer

Driven by real-time level feedback — no manual pumping

Working HMI

LCD status display and keypad input demonstrated

Integrated prototype

Mechanical, electrical, sensing, code and UI in one unit

07 ENGINEERING TAKEAWAY

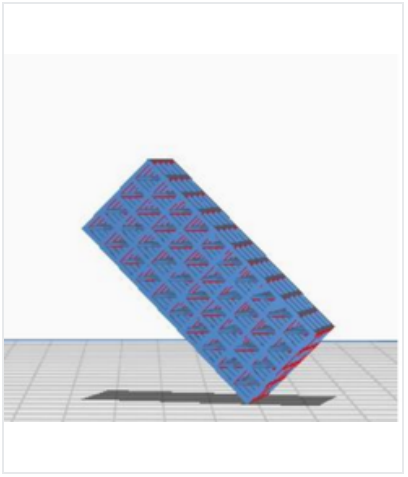
Mechatronic system integration — sensor, microcontroller, motor driver, power electronics and mechanical hardware working as one closed loop — plus the judgment to identify the least reliable component and target it for the next iteration.

P03 · NEW YORK UNIVERSITY

DMLS-Printed 316L Stainless-Steel Lattices — Tension & Compression Characterization

Designed, printed and tested 316L lattice structures to characterize stiffness, strength and failure behavior for lightweight, energy-absorbing applications.

DURATION	PROCESS	TESTING	ANALYSIS
Sep – Dec 2025	DMLS metal powder-bed fusion	Instron UTM · tension & compression	Force–displacement → engineering stress–strain



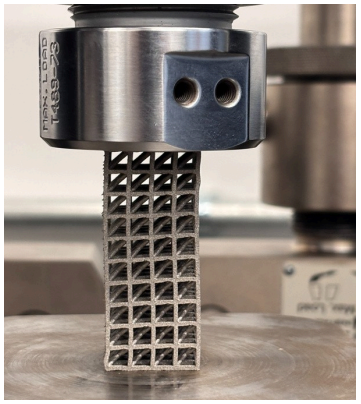
A Build orientation with supports — set before printing.



B Printed tensile specimen.



C Tensile test in progress.



D Compression test between UTM platens.



E Deformed lattice after loading — failure mode, not just peak load, is the result.

01 PROBLEM / OBJECTIVE

Metal AM lattices can make light, energy-absorbing structures — but their stiffness and strength depend on geometry and print quality. Goal: measure **stiffness, yield and ultimate strength** in tension and compression, and identify **how the lattices fail**.

02 ENGINEERING APPROACH — TEST WORKFLOW



03 DESIGN & DEVELOPMENT

Process decisions made to get repeatable specimens:

ORIENTATION	**45° (+X) and 15° (+Z)**
SCAN	Strategic scan patterns
SUPPORTS	Stabilized support structures
MATERIAL	SS316L powder, DMLS

05 TESTING & ITERATION

- **Tested** 5 tensile and 5 compression specimens on an Instron UTM.
- **Converted** raw force–displacement data into engineering stress–strain curves to compare samples.
- **Characterized** failure: **strut tearing** in tension; **progressive buckling** in compression.
- Compressive modulus varied (0.44–1.50 GPa) — attributed to lattice buckling, while strength stayed consistent.

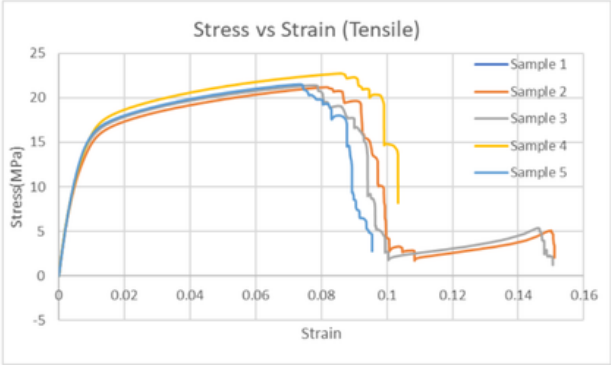


FIG 3.1 Tension, 5 samples: tight clustering to peak, then strut tearing drives the drop.

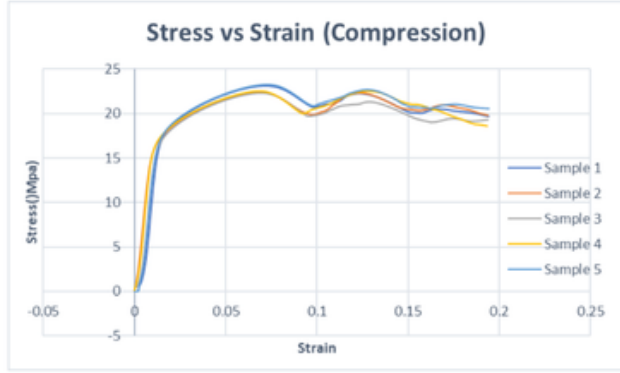


FIG 3.2 Compression, 5 samples: stress plateaus instead of dropping — progressive buckling, non-brittle.

06 RESULTS

LOAD CASE	STRENGTH	MODULUS	FAILURE MODE
Tension	**21–23** MPa UTS	**2.4–2.5** GPa	Strut tearing
Compression	**22–23** MPa	**0.44–1.50** GPa	Progressive buckling

Stable, repeatable tensile properties and a steady compressive strength confirm a **controlled, non-brittle failure profile** suited to energy-absorbing applications.

07 ENGINEERING TAKEAWAY

The full metal-AM qualification loop at small scale: build-process decisions, standardized mechanical testing, data reduction, and failure-mode interpretation — the same workflow used to decide whether an AM part is fit for service.

P04 · INDUS UNIVERSITY · B.TECH

Hydraulic Cross-Slide Retrofit for a Conventional Lathe

Designed and prototyped a semi-automated hydraulic cross-slide to replace a manual lathe mechanism — a low-cost alternative to CNC retrofitting.

DATE	TOOLS / SYSTEM
2023	Hydraulic cylinder · power pack · PLC control · CAD drawings

01 PROBLEM

Manual cross-feed demands constant operator effort and adds vibration and inconsistency to the cut.

02 APPROACH

Replace the handwheel-driven cross-slide with a **hydraulic cylinder and power pack**, mounted through a **custom plate**, with PLC-based control.

03 DESIGN

Designed and integrated the hydraulic cylinder, power pack and custom mounting plate, applying hydraulic principles.

05 TESTING

Tested facing, taper turning and grooving operations on the prototype.

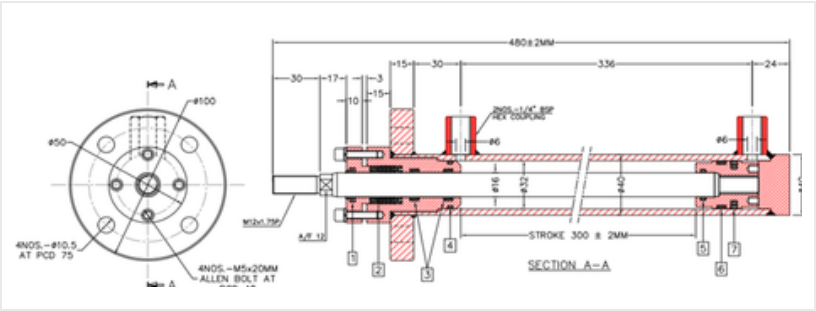


FIG 4.1 Cylinder end view and section A–A: 300 ± 2 mm stroke, 480 ± 2 mm overall length, port locations defined.



FIG 4.2 Cylinder installed on the lathe, driving the cross-slide.



FIG 4.3 Fabricated cylindrical component.



FIG 4.4 Hydraulic cylinder mounted on the lathe.



FIG 4.5 Custom mounting plate installed on the lathe.

06 RESULTS

~70%
Prototype success on trial operations

- Smoother cross-feed motion
- Reduced vibration and operator fatigue
- Improved surface finish
- Demonstrated a low-cost alternative to CNC retrofitting

07 ENGINEERING TAKEAWAY

Retrofitting a real machine: integrating fluid power, a custom mount and control onto existing equipment, then validating on actual cutting operations.

P05 · IACE GANDHINAGAR · ADVANCE CERTIFICATION

Injection Mold Design — Curtain Ring

Designed a complete injection mold in SolidWorks and gained hands-on time on an industrial injection molding machine.

DATE	TOOLS
2023 – 24	SolidWorks · FANUC Roboshot α-S100iA

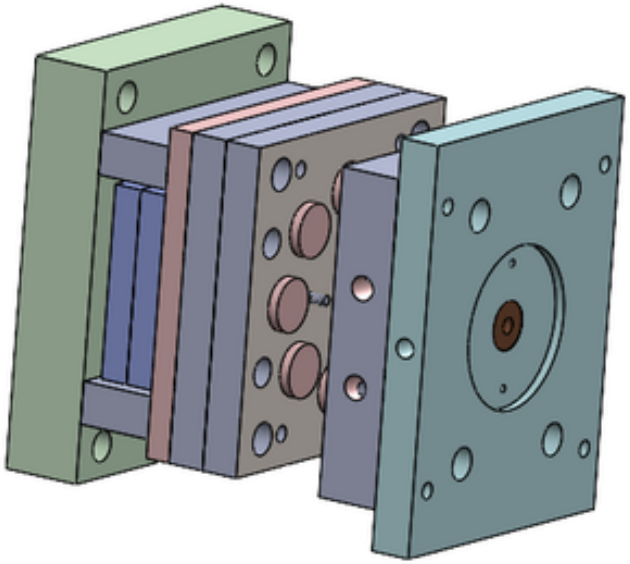


FIG 5.1 Exploded mold assembly — plate stack, core and cavity inserts, guide elements and fasteners designed as one system.

CREDENTIAL

Advance Certification Programme — Machining Technology & Plastic Engineering

International Automobile Centre of Excellence, Gandhinagar (JV of Govt. of Gujarat & Maruti Suzuki)

GRADE

Competent

Jun 2024

01 OBJECTIVE

Design a production-style mold for a curtain ring — every functional element a real tool needs, not just the part cavity.

03 MOLD ANATOMY DESIGNED

CORE & CAVITY · VENTING · EJECTION SYSTEM · COOLING CHANNELS · GUIDE PILLARS · BUSHES · FIXTURES · FASTENERS

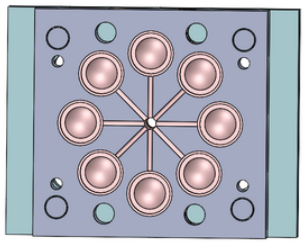


FIG 5.2 Eight-cavity layout fed by a radial runner from the central sprue.



FIG 5.3 Curtain rings — the part the mold is designed to produce.



FIG 5.4 FANUC Roboshot α-S100iA.

04 HANDS-ON

Gained hands-on experience on the FANUC Roboshot α-S100iA and hopper-dryer systems; studied plastic materials, part design, mold anatomy and mold-flow analysis in the program curriculum.

07 TAKEAWAY

Designing tooling forces DFM thinking: feeding, cooling and ejecting the part matter as much as its shape.

P06 · IACE GANDHINAGAR · ADVANCE CERTIFICATION

CNC / VMC Machining — From Drawing to Part

Designed parts, defined toolpaths, hand-programmed G-code and machined plastic, aluminum and copper-alloy components on FANUC-controlled CNC and VMC machines.

01 DESIGN Part drawing → **02 PLAN** Toolpaths & work-holding → **03 PROGRAM** Manual G-code → **04 MACHINE** FANUC CNC / VMC → **05 INSPECT** Calipers · micrometers

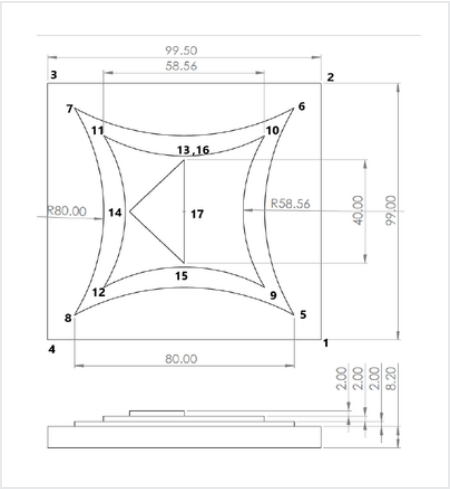


FIG 6.1 Dimensioned drawing with numbered profile points — the reference for toolpath coordinates.

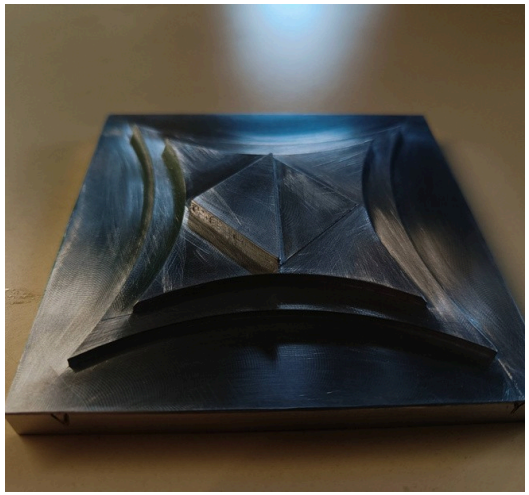


FIG 6.2 Machined part from that drawing: concave-arc profile, stepped pockets and a raised triangular feature.

03 WHAT I DID

- **Programmed** G-code by hand — linear and circular interpolation, canned cycles.
- **Applied** cutting parameters, machining process sequence and work-holding principles.
- **Machined** components from raw plastic, aluminum and copper-alloy stock, producing 3 validation samples per operation.
- **Inspected** parts with vernier calipers, micrometers and dial indicators; trained in GD&T.



FIG 6.3 Raw stock.

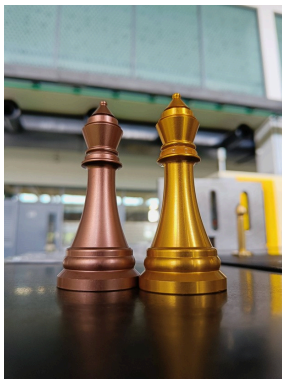


FIG 6.4 Turned profile parts.

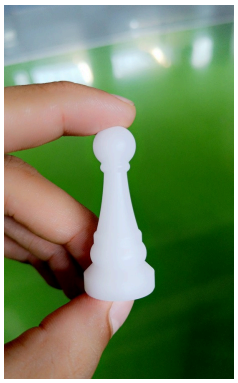


FIG 6.5 Turned profile part.



FIG 6.6 Multi-diameter turned part.

07 ENGINEERING TAKEAWAY

Knowing how a part is actually cut — toolpaths, fixturing, tolerances — makes every design I draw easier and cheaper to manufacture.

P07A · NYU MAKERSPACE · 2025

Mini CNC Plotter — 12-Hour Build

Designed and assembled a functional CNC plotter from scrap materials in under 12 hours.

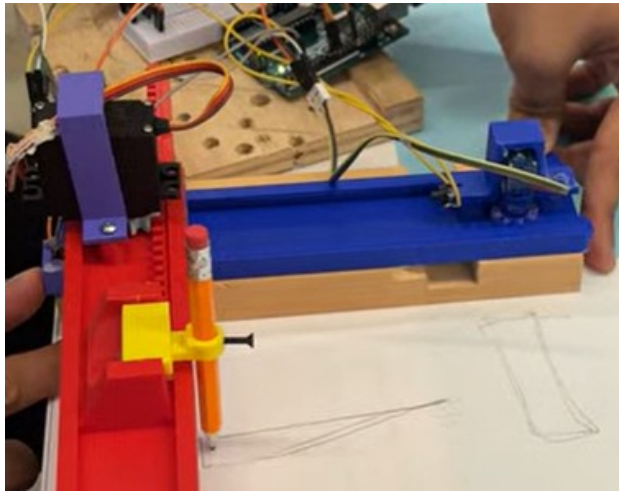


FIG 7.1 Servo-driven axes on a 3D-printed rack structure.

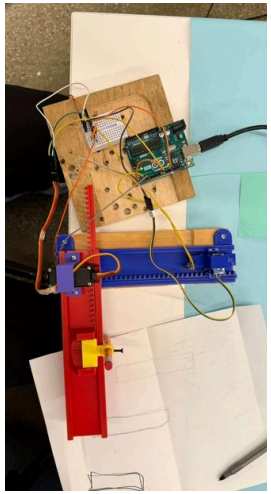


FIG 7.2 Arduino control on a scrap-wood base.

1st	<12h	$200
Place, NYU Makerspace Scrapathon	Concept to working machine	Grant for future projects

How: Arduino control, servo motors and a 3D-printed structure, built from scrap materials under a hard time limit.

— TAKEAWAY

Rapid prototyping under constraint: choose simple mechanisms, print what's missing, and get to a working machine fast.

P07B · NEW YORK UNIVERSITY · SEP – DEC 2025

Couette Flow — Thermal-Fluid Solver

Developed a Python finite-difference solver for coupled Navier–Stokes and energy equations, from incompressible to supersonic regimes.

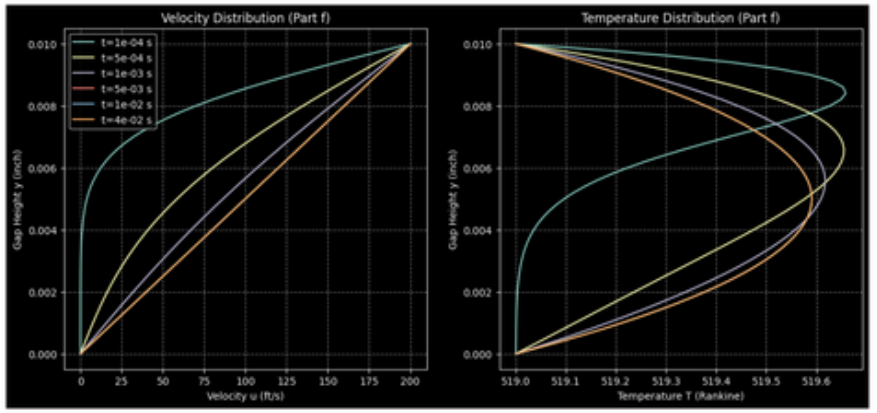


FIG 7.3 Transient velocity (left) and temperature (right) profiles across the gap as the flow develops.

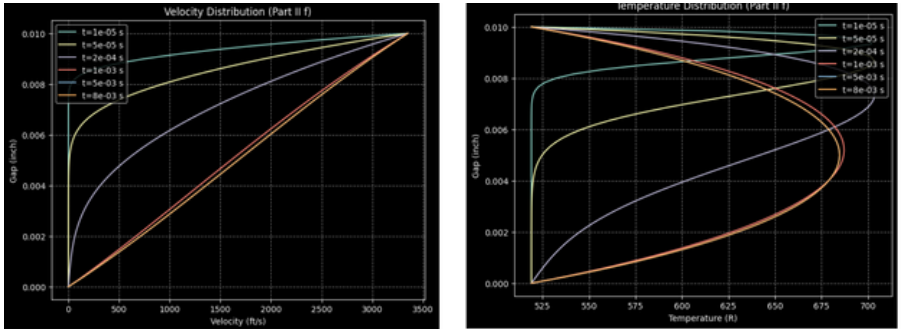


FIG 7.4 High-speed case: viscous dissipation produces the quadratic temperature rise.

521.316 °R	1.30×10^{-4}	−3.74
Adiabatic wall temp. at Ma ≈ 3.0	Skin-friction coefficient C_f	Viscous heat flux, BTU/s·ft²

Verified solution accuracy with grid-independence and time-step sensitivity studies.

— TAKEAWAY

Analysis to back up hardware: building and verifying my own numerical model rather than trusting a black box.

EXPERIENCE

Research Assistant — Computational Materials
JUN 2026 – NOW

New York University

Molecular-dynamics simulation of melting in Au, Al and Cu (LAMMPS / MedeA), validated against experimental data; built automated HPC workflows; mentor 2 undergraduate researchers.

Course Assistant — Materials Science Lab
SEP 2025 – NOW

New York University

Developed lab procedures and training for 150+ students (30+ per session) across XRD, SEM, tensile, hardness, metallography, heat treatment and laser cutting. NYU Lab Safety Certified.

Production Support Engineer
FEB – DEC 2024

GO3D Innovation LLP, Gandhinagar, India · see P01

Mechanical Engineering Intern
NOV 2023 – FEB 2024

GO3D Innovation LLP · DOE-style print testing, exhibitions, 5S / Kaizen · see P01

Advance Certification Trainee
2023 – 2024

IACE Gandhinagar · Machining Technology & Plastic Engineering · see P05–P06

Mechanical Engineering Intern — Quality
JUN – JUL 2021

Sunshine Enterprise, Ahmedabad, India

Quality inspections and dimensional verification with calipers, micrometers and gauges against ISO and customer requirements.

— SOFTWARE & PLATFORMS

SOLIDWORKS · CREO · AUTOCAD · CAD/CAM · MATERIALISE MAGICS · CURA · PYTHON · LAMMPS / MEDEA · PEEL 3D SCANNER · LPBF / DMLS · WAAM · FDM · SLA

— SOLIDWORKS PART MODELING



Part models built during SolidWorks training at IACE.

CAPABILITIES — MAPPED TO EVIDENCE

CAPABILITY	P01	P02	P03	P04	P05	P06	P07	NYU
Mechanical design & CAD (SolidWorks)	●	–	●	●	●	●	●	–
Engineering drawings & GD&T	●	–	–	●	–	●	–	–
DFM / DFA	●	–	–	–	–	–	–	–
Polymer AM — FDM / resin	●	●	–	–	–	–	●	–
Metal AM — DMLS build setup	–	–	●	–	–	–	–	–
Prototyping & hands-on assembly	●	●	–	●	–	–	●	–
Mechanical testing (UTM)	–	–	●	–	–	–	–	●
Materials characterization (SEM, XRD, hardness)	–	–	–	–	–	–	–	●
Embedded control (STM32 / Arduino)	–	●	–	–	–	–	●	–
Motors & drivers (stepper, servo, TMC2209)	–	●	–	–	–	–	●	–
Sensors & closed-loop control	–	●	–	–	–	–	–	–
Power electronics & wiring	–	●	–	–	–	–	–	–
Hydraulics & PLC control	–	–	–	●	–	–	–	–
Injection molding & tooling	–	–	–	–	●	–	–	–
CNC programming (G-code) & metrology	–	–	–	–	–	●	–	–
Calibration & root-cause analysis	●	●	●	–	–	–	–	–
Lean / 5S / Kaizen process improvement	●	–	–	–	–	–	–	–
Numerical modeling (Python, MD)	–	–	–	–	–	–	●	●
Technical training & SOPs	●	–	–	–	–	–	–	●

● applied in this project · NYU = Research & Course Assistant roles



LET'S BUILD SOMETHING

Vedant Nadoda

Mechanical Engineer · M.S. New York University, December 2026

Seeking full-time roles in mechanical, additive manufacturing, manufacturing and applications engineering, product development, R&D / prototyping, and mechatronics.

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